

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
FSDT Midterm Examinations Spring 2026

Time: 90 minutes

CT353 – Operating System Paper A

Max Marks: 20

Note: Attempt all questions.

Question 1 [CLO 2 – 10 Marks]

- a. Compare the direct and indirect modes of IPC message passing.
- b. Discuss how computers handle data transfer (I/O) in operating systems.
- c. Illustrate what an API is and discuss the steps involved in its implementation.
- d. Infer process states with the help of a diagram.
- e. Illustrate how the operating system manages data protection and security with examples.

Question 2 [CLO 4 – 10 Marks]

- a. Analyze process scheduling and its types. Also, illustrate the difference between I/O-bound and CPU-bound processes.
- b. Examine the PCB and diagrammatically show how the CPU switches from one process to another.
- c. Assume a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. The number of frames in memory is 3. Find out the number of page faults with respect to:
 - i. Optimal Page Replacement Algorithm
 - ii. FIFO Page Replacement Algorithm